Day 1 Notes Computer Networks

Day 1: CN Basics + OSI + TCP/IP

Topics covered:

- What is Computer Network
- Types of Networks: LAN, MAN, WAN
- Topologies: Bus, Star, Ring, Mesh
- OSI Model
- TCP/IP Model
- Difference between OSI and TCP/IP
- Encapsulation and Decapsulation

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1. What is a Computer Network

A **computer network** is a collection of two or more computers/devices connected together so that they can **communicate**, **share resources**, and **exchange data**.

These devices may include:

- Computers
- Mobile phones

- Servers
- Printers
- Routers
- Switches
- IoT devices

The connection can be made through:

- **Wired medium**: copper cable, fiber optic cable
- **Wireless medium**: Wi-Fi, Bluetooth, satellite, radio waves

Why do we need computer networks?

Computer networks are needed because they make communication and resource sharing easier.

Main purposes:

1. **Resource sharing**
    - Share printer, files, internet connection, storage
2. **Communication**
    - Email, chat, video call, messaging
3. **Data sharing**

    - Transfer files, access shared databases

4. **Remote access**
    - Access a system from another location
5. **Centralized management**
    - Control data, users, and security from one place
6. **Scalability**
    - Add more devices easily
7. **Reliability**
    - Backup and redundancy can reduce failures

Example

If a college has:

- one central server,
- many student computers,
- Wi-Fi access,
  then all of them together form a network.

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2. Basic Components of a Computer Network

A network generally consists of the following parts:

1. Sender

The device that sends data.

2. Receiver

The device that receives data.

3. Transmission Medium

The path through which data travels.
Examples:

- Twisted pair cable
- Coaxial cable
- Fiber optics
- Radio waves

4. Protocol

A set of rules used for communication.

Examples:

- HTTP

- TCP
- IP
- FTP
- SMTP

5. Network Devices

- **Hub**
- **Switch**
- **Router**
- **Modem**
- **Access Point**
- **Repeater**
- **Bridge**
- **Gateway**

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3. Characteristics of a Good Network

A good network should have:

1. Performance

It should be fast and efficient.

2. Reliability

It should work continuously with minimum failure.

3. Security

It should protect data from unauthorized access.

4. Scalability

It should allow addition of more users/devices.

5. Manageability

It should be easy to configure and maintain.

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4. Types of Networks

Networks are classified mainly on the basis of geographical coverage.

A. LAN – Local Area Network

A **LAN** is a network that covers a **small geographical area**.

Examples:

- Home Wi-Fi

- School lab
- Office building
- College campus room or department

Features:

- Small area
- High speed
- Low cost
- Privately owned
- Easy to manage

Example

A computer lab in a college where 20 computers are connected to one switch.

Advantages:

- Easy file and printer sharing
- Fast communication
- Lower setup cost
- Better control over data

Disadvantages:

- Limited area
- Breakdown affects local users
- Needs administration

B. MAN – Metropolitan Area Network

A **MAN** covers a **city or large town**.

It is bigger than LAN but smaller than WAN.

Examples:

- Cable TV network across a city
- City-wide internet network
- Branches of a bank within the same city

Features:

- Medium geographical coverage
- Higher cost than LAN
- Connects multiple LANs
- Often managed by government/private companies

Advantages:

- Connects multiple offices in a city
- Good speed
- Useful for large organizations

Disadvantages:

- More expensive than LAN
- More complex management
- Security is harder than LAN

C. WAN – Wide Area Network

A **WAN** covers a **very large area**, such as countries or continents.

Example:

- The Internet
- Multinational company network across different countries

Features:

- Very large coverage
- Uses leased lines, satellites, public networks
- Higher latency than LAN

- Expensive and complex

Advantages:

- Global communication
- Connects distant branches
- Supports remote work

Disadvantages:

- High cost
- Slower than LAN
- Security issues
- Difficult to maintain

Difference between LAN, MAN, WAN

Feature	LAN	MAN	WAN
Full form	Local Area Network	Metropolitan Area Network	Wide Area Network
Coverage	Small area	City/town	Country/continent/world
Speed	High	Medium to high	Lower than LAN
Cost	Low	Moderate	High

Ownership	Private	Private/Public	Mostly public/private mix
Example	School lab	City bank branches	Internet

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5. Network Topology

Topology means the physical or logical arrangement of devices in a network.

In simple words, it tells us **how devices are connected**.

There are many network topologies, but the important ones here are:

- Bus
- Star
- Ring
- Mesh

A. Bus Topology

In **bus topology**, all devices are connected to a **single main cable**, called the **backbone**.

Working:

When a device sends data, the data travels through the backbone. All devices receive it, but only the destination accepts it.

Diagram idea:

`PC1 — PC2 — PC3 — PC4` on one common cable

Advantages:

- Easy to install
- Low cost
- Requires less cable
- Good for small networks

Disadvantages:

- If backbone fails, whole network fails
- Difficult to troubleshoot
- Performance decreases with more devices
- Data collision may happen

Use case:

Used in early Ethernet networks.

B. Star Topology

In **star topology**, all devices are connected to a **central device**, usually a **switch** or **hub**.

Working:

If one computer wants to send data to another, it sends it to the central switch, which forwards it to the destination.

Advantages:

- Easy to install and manage
- Easy fault detection
- One cable failure affects only one device
- Better performance

Disadvantages:

- If central device fails, whole network fails
- Requires more cable
- Slightly higher cost

Use case:

Most modern LANs use star topology.

C. Ring Topology

In **ring topology**, each device is connected to **two neighboring devices**, forming a ring.

Working:

Data moves in one direction (or both directions in some systems) around the ring until it reaches the destination.

Advantages:

- No data collision in controlled ring systems
- Equal access for all devices

Disadvantages:

- Failure of one device/cable can affect network
- Difficult to add/remove devices
- Troubleshooting is harder

Use case:

Older token ring networks.

D. Mesh Topology

In **mesh topology**, devices are connected to **many or all other devices**.

Types:

1. **Full Mesh** – every device connected to every other device
2. **Partial Mesh** – some devices connected to multiple devices

Advantages:

- Very reliable

- High fault tolerance
- Multiple paths available
- Better security and redundancy

Disadvantages:

- Very expensive
- Requires lots of cables and ports
- Complex installation and maintenance

Use case:

Military systems, backbone networks, critical systems

Comparison of Topologies

Topology	Structure	Cost	Reliability	Easy to Maintain
Bus	Single backbone	Low	Low	Difficult
Star	Central device	Medium	Good	Easy
Ring	Circular connection	Medium	Medium	Moderate
Mesh	Multiple interconnections	High	Very high	Difficult

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6. OSI Model

The **OSI Model** stands for **Open Systems Interconnection Model**.

It is a **7-layer conceptual model** used to explain how data moves from one device to another over a network.

It was developed by **ISO**.

The main purpose of the OSI model is to:

- standardize network communication
- make troubleshooting easier
- divide communication into layers
- allow interoperability between vendors

7 Layers of OSI Model

From top to bottom:

1. Application
2. Presentation
3. Session
4. Transport
5. Network
6. Data Link

7. Physical

A memory trick:
All People Seem To Need Data Processing

Layer 7: Application Layer

This is the **topmost layer** and closest to the user.

Function:

Provides network services directly to end-user applications.

Examples:

- HTTP / HTTPS
- FTP
- SMTP
- POP3
- IMAP
- DNS

Real-life meaning:

When you open a browser and access a website, this layer is involved.

Responsibilities:

- File transfer
- Email services
- Web browsing
- Remote login

Layer 6: Presentation Layer

This layer is responsible for the **format of data**.

Function:

It ensures that data sent by one system can be understood by another.

Responsibilities:

- Translation
- Encryption
- Decryption
- Compression
- Data formatting

Example:

- Converting text into ASCII/Unicode

- SSL/TLS encryption conceptually fits here in many textbook explanations

Real-life meaning:

It acts like a translator between systems.

Layer 5: Session Layer

This layer is responsible for **establishing, managing, and terminating sessions**.

Responsibilities:

- Session establishment
- Session maintenance
- Session termination
- Dialog control
- Synchronization

Example:

When two systems communicate continuously in a login session.

Real-life meaning:

It keeps track of whose turn it is to communicate and maintains the connection session.

Layer 4: Transport Layer

This layer provides **end-to-end delivery** of data.

Responsibilities:

- Segmentation
- Flow control
- Error control
- Reliable delivery
- Port addressing

Main protocols:

- **TCP** – reliable, connection-oriented
- **UDP** – faster, connectionless, less reliable

Key concepts:

- Segments
- Port numbers
- Acknowledgment
- Retransmission

Real-life meaning:

It ensures complete and correct delivery of data.

Layer 3: Network Layer

This layer is responsible for **logical addressing** and **routing**.

Responsibilities:

- Path selection
- Routing
- Logical addressing (IP address)
- Packet forwarding

Protocols:

- IP
- ICMP
- ARP (often discussed around this area)
- Routing protocols

Devices:

- Router

Data unit:

- Packet

Real-life meaning:

It decides which path data should take to reach destination.

Layer 2: Data Link Layer

This layer provides **node-to-node delivery**.

Responsibilities:

- Framing
- Physical addressing (MAC address)
- Error detection
- Flow control
- Access to media

Sub-layers:

1. LLC – Logical Link Control
2. MAC – Media Access Control

Devices:

- Switch

- Bridge

Data unit:

- Frame

Real-life meaning:

It ensures data reaches the correct device on the same local network.

Layer 1: Physical Layer

This is the **lowest layer**.

Responsibilities:

- Transmission of raw bits
- Defines cables, connectors, voltage, timing, data rate

Examples:

- Ethernet cables
- Fiber optic cables
- Radio signals
- Hubs, repeaters

Data unit:

- Bits

Real-life meaning:

It deals with actual hardware transmission.

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7. Summary Table of OSI Layers

Layer No.	Layer Name	Main Function	Data Unit	Device/Example
7	Application	User services	Data	HTTP, FTP
6	Presentation	Translation, encryption	Data	SSL format conversion
5	Session	Session management	Data	Session control
4	Transport	End-to-end delivery	Segment	TCP, UDP
3	Network	Routing, IP addressing	Packet	Router, IP
2	Data Link	MAC addressing, framing	Frame	Switch
1	Physical	Raw transmission	Bits	Cable, Hub

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8. TCP/IP Model

The **TCP/IP model** is the practical model used on the Internet.

It stands for:

- **TCP** = Transmission Control Protocol
- **IP** = Internet Protocol

This model is simpler and more practical than OSI.

Layers of TCP/IP Model

There are generally **4 layers**:

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Network Access Layer

Some books mention 5 layers by separating physical and data link, but the standard textbook comparison usually uses 4 layers.

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1. Application Layer

This combines the functions of:

- OSI Application
- OSI Presentation

- OSI Session

Responsibilities:

- User interaction
- Data formatting
- Communication services

Protocols:

- HTTP
- HTTPS
- FTP
- SMTP
- DNS
- Telnet

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2. Transport Layer

Same purpose as OSI transport layer.

Responsibilities:

- End-to-end communication

- Segmentation
- Error control
- Reliable or unreliable delivery

Protocols:

- TCP
- UDP

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3. Internet Layer

This corresponds roughly to the **Network layer** of OSI.

Responsibilities:

- Logical addressing
- Routing
- Packet forwarding

Protocols:

- IP
- ICMP
- ARP

- IGMP

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4. Network Access Layer

This combines:

- OSI Data Link layer
- OSI Physical layer

Responsibilities:

- Physical transmission
- Framing
- MAC addressing
- Hardware addressing
- Access to media

Examples:

- Ethernet
- Wi-Fi
- Token Ring

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9. TCP/IP Layer Table

TCP/IP Layer	Main Function	Related Protocols
Application	User-level communication	HTTP, FTP, SMTP, DNS
Transport	End-to-end communication	TCP, UDP
Internet	Routing and IP addressing	IP, ICMP, ARP
Network Access	Local delivery and physical transmission	Ethernet, Wi-Fi

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10. Difference between OSI and TCP/IP Model

This is a very important theory question.

Basis	OSI Model	TCP/IP Model
Full form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Number of layers	7	4
Type	Reference model	Practical protocol suite
Developed by	ISO	DARPA/DoD
Usage	Mainly conceptual and educational	Used in real-world internet

Session and Presentation	Separate layers	Included in Application layer
Physical and Data Link	Separate layers	Combined into Network Access
Protocol dependency	Protocol independent	Protocol oriented
Complexity	More detailed	Simpler

Important Points

1. **OSI is theoretical**, TCP/IP is **practical**.
2. **OSI has 7 layers**, TCP/IP has **4 layers**.
3. **TCP/IP is used on the Internet**.
4. In TCP/IP, upper 3 OSI layers are combined into one Application layer.